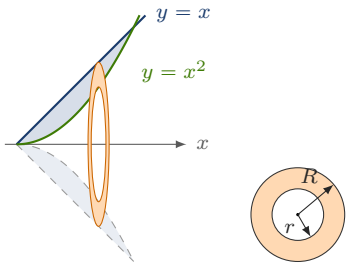
L4.2b Washers

Washers and rotating about other lines — what you need to know

WASHERS



When the region does not touch the axis, the solid has a hole down the middle, and every slice is a disk with a hole punched in it: a **washer**.

$$A(x) = \pi R(x)^2 - \pi r(x)^2$$

$R(x)$ is the outer radius, $r(x)$ the inner radius. The washer of thickness dx is one little chunk of the solid, $dV = A(x)\,dx$. A region that touches the axis gives a disk: a washer with no hole.

"the disk out to the outer edge, minus the disk out to the inner edge"

VOLUME BY WASHERS

① **Draw a representative washer:** a thin rectangle **perpendicular** to the spin axis, from the inner curve to the outer curve, and the washer dV it sweeps. Label both radii from the axis: R to the outer curve, r to the inner curve.

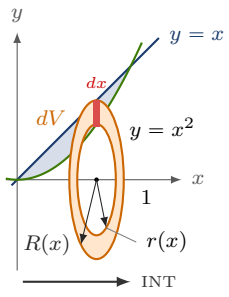
② **Path of integration and limits:** along the spin axis — dx for a horizontal axis, dy for a vertical one — from where the rectangles start to where they stop.

③ **Formulas for the radii,** $R(x)$ and $r(x)$ or $R(y)$ and $r(y)$: how far from the axis to each curve, in the variable of the path. Measure from the axis, not from 0.

④ **The chunk:** $dV = A(x)\,dx = \left(\pi R(x)^2 - \pi r(x)^2\right) dx$, or the same in y.

$$V = \int_{\text{path}} dV$$

WASHERS ABOUT THE x-AXIS



Ex. The region enclosed by $y = x$ and $y = x^2$ is rotated about the x-axis. Find the volume.

① **Draw a representative washer:** an upright rectangle from the inner curve up to the outer curve, the washer dV, and its radii $R(x)$, $r(x)$.

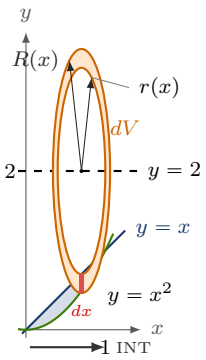
② **Path and limits:** dx, from 0 to 1, where the curves cross.

③ **Radii:** test $x = \frac{1}{2}$: $\frac{1}{2} > \frac{1}{4}$, so $y = x$ is the outer curve: $R(x) = x$, $r(x) = x^2$.

④ **Chunk:** $dV = A(x)\,dx = \pi\left(x^2 - x^4\right) dx$.

$$V = \int_{\text{path}} dV = \pi \int_0^1 x^2 - x^4\,dx = \pi\left[\frac{1}{3} - \frac{1}{5}\right] = \boxed{\frac{2\pi}{15}}$$

WASHERS ABOUT A HORIZONTAL LINE



Ex. The same region, rotated about the line $y = 2$.

① **Draw a representative washer:** an upright rectangle, the washer dV it sweeps about $y = 2$, and $R(x)$, $r(x)$. The axis is **above** the region, so the outer curve is now the lower one, $y = x^2$.

② **Path and limits:** dx, from 0 to 1.

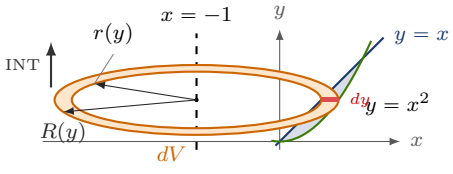
③ **Radii:** measure R and r from $y = 2$: $R(x) = 2 - x^2$, $r(x) = 2 - x$.

④ **Chunk:** $dV = A(x)\,dx = \left(\pi(2 - x^2)^2 - \pi(2 - x)^2\right) dx$.

$$A(x) = \pi\left(4 - 4x^2 + x^4 - 4 + 4x - x^2\right) = \pi\left(x^4 - 5x^2 + 4x\right)$$

$$V = \int_{\text{path}} dV = \pi \int_0^1 x^4 - 5x^2 + 4x\,dx = \pi\left[\frac{1}{5} - \frac{5}{3} + 2\right] = \boxed{\frac{8\pi}{15}}$$

WASHERS ABOUT A VERTICAL LINE



Ex. The same region, rotated about the line $x = -1$.

① **Draw a representative washer:** a sideways rectangle from the inner curve $y = x$ across to the outer curve $y = x^2$, the washer dV it sweeps about $x = -1$, and $R(y)$, $r(y)$.

② **Path and limits:** dy, from 0 to 1.

③ **Radii:** how far from $x = -1$ to the curve: $1 + x$, but needed in terms of y. Inner curve $y = x$: $r(y) = 1 + y$. Outer curve $y = x^2$, $x = \sqrt{y}$: $R(y) = 1 + \sqrt{y}$.

④ **Chunk:** $dV = A(y)\,dy = \pi\left[\left(1 + \sqrt{y}\right)^2 - (1 + y)^2\right] dy$.

$$A(y) = \pi\left[1 + 2\sqrt{y} + y - 1 - 2y - y^2\right] = \pi\left(2\sqrt{y} - y - y^2\right)$$

$$V = \int_{\text{path}} dV = \pi \int_0^1 2\sqrt{y} - y - y^2\,dy = \pi\left(\frac{4}{3} - \frac{1}{2} - \frac{1}{3}\right) = \boxed{\frac{\pi}{2}}$$

WATCH OUT

✗ ~~$A = \pi(R - r)^2$~~ → $A = \pi R^2 - \pi r^2$ — **square each radius, then subtract**

✗ ~~about $y = 2$, the radius out to $y = x^2$ is x^2~~ → **measure from the axis:** $2 - x^2$

✗ ~~x-limits on a dy integral~~ → **the limits are the y-coordinates of the crossings**